

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

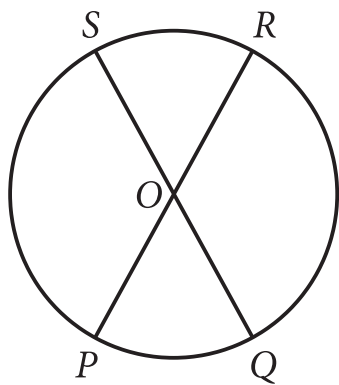
Geometry and Trigonometry

06

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

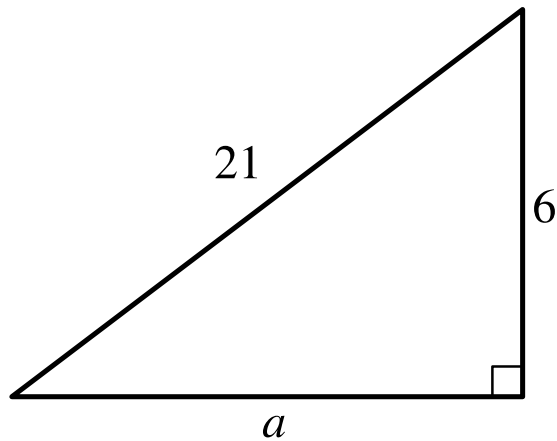


Note: Figure not drawn to scale.

- The center of the circle is point O .
- Points S , R , Q , and P are on the circle.
- Line segment PR is a diameter of the circle.
- Line segment QS is a diameter of the circle.
- Diameters PR and QS intersect at point O .
- A note indicates the figure is not drawn to scale.

The circle shown has center O , circumference 144π , and diameters $\overline{PR}$ and $\overline{QS}$. The length of arc PS is twice the length of arc PQ . What is the length of arc QR ?

- A. 24π
- B. 48π
- C. 72π
- D. 96π

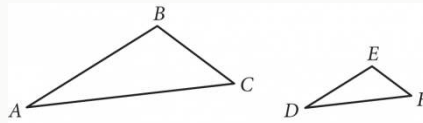


Note: Figure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is 21.
- The length of one side adjacent to the right angle is 6.
- The length of the other side adjacent to the right angle is a .
- A note indicates the figure is not drawn to scale.

For the triangle shown, which expression represents the value of a ?

- A. $\sqrt{21^2 - 6^2}$
- B. $21^2 - 6^2$
- C. $\sqrt{21 - 6}$
- D. $21 - 6$



Note: Figures not drawn to scale.

Triangle ABC and triangle DEF are shown. The relationship between the side lengths of the two triangles is such that $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = 3$. If the measure of angle BAC is 20° , what is the measure, in degrees, of angle EDF ? (Disregard the degree symbol when gridding your answer.)

STUDENT-PRODUCED RESPONSE

○	/○	/○	○
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4GEOMETRY AND TRIGONOMETRY

A circle has a radius of 43 meters. What is the area, in square meters, of the circle?

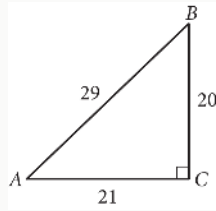
- A. $\frac{43\pi}{2}$
- B. 43π
- C. 86π
- D. $1,849\pi$

Q5GEOMETRY AND TRIGONOMETRY

$$x^2 + 58x + y^2 = 0$$

In the xy -plane, the graph of the given equation is a circle. What are the coordinates x, y of the center of the circle?

- A. 0, 29
- B. 0, - 29
- C. 29, 0
- D. -29, 0



In the figure above, what is the value of $\tan(A)$?

- A. $\frac{20}{29}$
- B. $\frac{21}{29}$
- C. $\frac{20}{21}$
- D. $\frac{21}{20}$

Right triangles LMN and PQR are similar, where L and M correspond to P and Q , respectively. Angle M has a measure of 53° . What is the measure of angle Q ?

- A. 37°
- B. 53°
- C. 127°
- D. 143°

Q8

GRID-IN

GEOMETRY AND TRIGONOMETRY

A triangular prism has a height of 8 centimeters cm and a volume of 216 cm^3 . What is the area, in cm^2 , of the base of the prism? (The volume of a triangular prism is equal to Bh , where B is the area of the base and h is the height of the prism.)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN

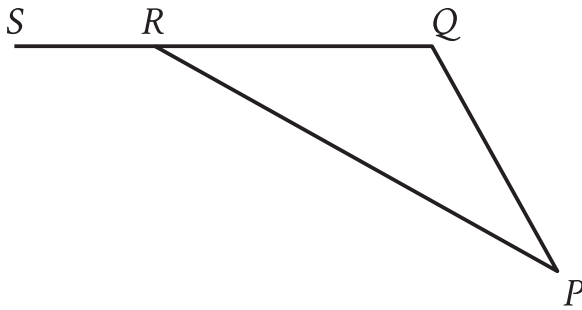
GEOMETRY AND TRIGONOMETRY

An angle has a measure of $\frac{9\pi}{20}$ radians. What is the measure of the angle in degrees?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Line segment QR is extended to point S .
- A note indicates the figure is not drawn to scale.

In triangle PQR , QR is extended to point S . The measure of $\angle PQR$ is 132° , and the measure of $\angle PRS$ is 163° . What is the measure of $\angle QPR$?

- A. 48°
- B. 31°
- C. 24°
- D. 17°